

B. Tech Degree V Semester Examination November 2010

ME 502 METROLOGY AND MACHINE TOOLS

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer all questions)

(8 x 5 =40)

- I. (a) Explain line standards and end standards.
- (b) Give the classification of gauges.
- (c) Outline the CLA method of assessment of surface finish.
- (d) Discuss the role of optical interference in metrology.
- (e) Explain drilling and boring. What is trepanning?
- (f) Write briefly on Capstan and Turret lathes.
- (g) Distinguish between upmilling and downmilling with neat diagram.
- (h) Elucidate chipless machining process.

PART B

(4 x 15 =60)

- II. Explain sensitivity, hysteresis, response time, repeatability, precision, accuracy and traceability.

OR

- III. Discuss limits, fits and tolerance in engineering metrology.

- IV. With sketches describe angular measurement of instruments.

OR

- V. With diagram describe optical measuring instruments.

- VI. Give the classification of lathes. Describe with diagram any one of them.

OR

- VII. Illustrate the application of quick return motion mechanism in machine tools.

- VIII. Explain grinding, boring, lapping, gear cutting and gear shaping operators with drawings.

OR

- IX. Describe Numerical Control Machine Tools and automatic machines.